International Trade and Credit Market Imperfection

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We introduce credit market imperfection in an otherwise Ricardian model, but for identical technologies, when workers or entrepreneurs are endowed with different levels "capital" and need to borrow to produce the credit intensive good. We argue that in such a framework identical countries will gain from trade without the assumption of comparative advantage. Thus, we show that we do not need monopolistically competitive models to generate trade between similar countries. International trade, in our set up, takes place in fragments in a way that can help alleviate credit constraints. Moreover, very poor firms and very rich ones are not likely to gain from trade in fragments, but those in the middle will.

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I. Introduction

Do credit market imperfections create incentive to engage in international trade? In other words, do identical countries characterized by imperfection in credit market gain from trading with each other? This paper aims at addressing this question. Our results support the conclusion that under such circumstances, gains from trade accrue even if the countries do not enjoy comparative advantage of any kind. In a way we provide a Ricardian analogue to the Dixit-Stiglitz-Krugman (DSK) variety model of trade [Dixit and Stiglitz (1977), Krugman(1990)]. Trade in the DSK genre of models leads to gain not because countries enjoy comparative advantage in producing a good or variety of goods, but because trade allows greater number of varieties to be consumed. Thus, trade can take place between identical countries negating the asymmetry that is essential in more conventional models such as in Ricardian or Heckscher-Ohlin-Samuelson type models.

We demonstrate how identical nations can gain from trade when they suffer from credit market imperfections. Thus one can coin our model as one which deals with "South - South Trade". But this is not the only way our result can be interpreted. If a country without having credit constraints trades with another which is affected by the shortage of credit, it is likely that the credit constrained economy will specialize in fragments. In this case there is a comparative advantage story at work, the credit constrained

economy has a comparative advantage in supplying fragments. But the bottom line is that trade is perfectly feasible between countries which face such constraints and they can be identical in all respects. Let us, at the outset, draw a distinction between our work and the landmark papers by Matsuyama (2005) and Deardorff (2000) that have brought to the fore the issue of credit and finance in the context of international trade and comparative advantage. Our paper is distinct from both in that we explicitly focus on credit market imperfection and fragmentation. On the one hand, in comparison with Matsuyama (2005), our model is not necessarily tied to comparative advantage. On the other hand, Deardroff (2000) argues how, during financial crises, fragmentation (i.e. splitting up of production process in various traded segments) may create problems. We show, instead, how the ability to trade in fragments can help alleviate credit constraints.

Our paper also shows how trade promotes entrepreneurs and how trade actually helps the poor people more in relation to wealthier ones i.e. those endowed with higher amounts of capital. As such, our work is related to Jones and Marjit (2001) where fragmentation has been shown to help the young generation relative to the old who tend to own capital. Meisenzahl (2011), in a comprehensive study on whether financing constraint actually determines firm size, shows that constrained firms do have smaller size and younger firms are relatively small compared to older firms as the older ones are likely to get better credit facilities. Such a young-old asymmetry not only corroborates Jones and Marjit (2001) conjecture, but also lends credence to the theoretical model under consideration where we show that constrained entrepreneurs will opt for smaller size if they can. In Marjit (2008), relative capital endowments determine pattern of trade and occupational pattern simultaneously in a Heckscher-Ohlin setting, but the main focus was on wage distribution. Lineage wise this paper is related to papers dealing with more 'classical' interpretation of the Ricardian theory of comparative advantage a' la Steedman (1979) and Findlay (1984) and also, though tangentially, to the literature on north-south trade [Findlay (1995)].

While we produce a clear result where exactly similar countries gain from trade, a reinterpreted version in the context of the 'north-south' trade also yields an interesting result. The fact that a country faces credit market imperfection while the other does not, implies that the credit constrained economy

will have a comparative advantage in producing fragments. Such a trade pattern echoes the earlier concern of trade between metropolis and colonies popularized through the literature on trade between centre and periphery. Two earlier attempts in modeling such a behavior without including capital are by Sanyal (1983) and Marjit (1987). The rest of the paper is organized as follows. Section 2 describes the model and presents the key results. Sections 3 and 4 look into further consequences of our results. In the final section, we draw our key conclusions.

II. Model

Consider an economy producing two goods – x and y. x is produced with labor: 'a' units of labor produces one unit of x. Production of y requires one period to complete and workers have to be paid at the beginning, thus entrepreneurs require credit for the wage fund. One unit of labor produces one unit of y. The distribution is given in a continuum such that each worker is represented by $z \epsilon [0,1]$. Each worker is endowed with a wage fund $k(z)$ called 'capital'. In effect, we stylize is a Ricardian small open economy, where workers are entrepreneurs, with commodity prices and interest rates determined in the rest of the world. However, the credit market is imperfect. If the workers so wish they can buy a bond which pays them r, a return as a lender determined in the rest of the world. Thus, so far as the lending or deposit rates goes, it is fixed at r. However, if they wish to borrow to augment their capital stock they must pay $R > r$. In the standard literature on credit market imperfection such divergence is a common occurrence. Such a summary view of the imperfect credit market has been quite popular in theories of occupational distribution and poverty trap, as surveyed in Basu (2006). One can easily bring in adverse selection or moral hazard type assumptions to create an endogenous wedge between R and r. But that is not necessary for the purpose in hand.

At this stage, we make an assumption that $k(z)$ is small enough for all z such that it falls short of the required wage fund to produce y. Since to each worker-entrepreneur the imputed cost of producing is $\frac{1}{a}$, the alternative wage rate, our assumption suggests that $k(z) < \frac{1}{a}\ \forall\ z$. Once they decide to borrow from

the international market, the banks have to incur a loan processing cost, say, $\alpha > 1$ in addition to r, the lending rate. Thus $R = \alpha r > r$ As we shall see shortly given that $R > r$, internal source of financing such projects becomes crucial as predicted in the standard finance literature [Glenn Hubbard (1989)]. We also assume competitive commodity markets where price has to be equal to average cost of production.

With this backdrop, we can now specify the incentive of a representative worker to be an entrepreneur and engage in producing y rather than x.

$$P + \frac{1}{a}(1+r) - \left[\frac{1}{a} - k(z)\right](1+R) \geq \left[\frac{1}{a} + k(z)\right](1+r) \tag{1}$$

where $P \equiv \frac{P_y}{P_x}$.

It may be noted that RHS represents the opportunity cost of becoming an entrepreneur and (1) can be rewritten as

$$P + k(z)(R-r) \geq \frac{1}{a}(1+R) \tag{2}$$

In the absence of any imperfections in the credit market, $r = R$. Hence, (2) boils down to

$$P \geq \frac{1}{a}(1+r) \tag{3}$$

Internal financing becomes irrelevant as $k(z)$ vanishes from the scene with $r = R$.

(3) also suggests that whether the economy will specialize in x or y depend on whether the return from entrepreneurship $\left[P - \frac{1}{a}(1+r)\right]$ is positive.

Complete specialization will be the likely outcome if $> \frac{1}{a}(1+r)$, replicating the fundamental Ricardian proposition.

Let us now rank $k(z)$ such that $k'(z) > 0$ for $z \epsilon [0,1]$ with $P + k(1)(R-r) > \frac{1}{a}(1+R)$ and $P + k(0)(R-r) < \frac{1}{a}(1+R)$ and define $\tilde{z}$ such that

$$P + k(\tilde{z})(R-r) = \frac{1}{a}(1+R) \tag{4}$$

It is obvious from (3) and (4) that since $r < R$, any incentive to produce y is greater with a distorted credit market. However the net incentive, if any, to engage in production of y will be greater in an undistorted credit market, since the following holds

$$\left[P - \frac{1}{a}(1+r)\right] - \left[P + k(z)(R-r) - \frac{1}{a}(1+R)\right] = \left[\frac{1}{a} - k(z)\right](R-r) > 0 \; as \; k(z) \leq \frac{1}{a} \; for \; z \leq \tilde{z} \, .$$

Since $k'(z) > 0, \; \forall \; z \geq \tilde{z}$ workers will become entrepreneurs in y and for the rest, x seems to be more profitable. Thus even if (3) holds with strict inequality, (4) may discourage all $z < \tilde{z}$ to become entrepreneurs in y, thus causing deviation from the first best allocation.

Therefore $(1 - \tilde{z})$ will be the amount of y that will be produced and $\frac{\tilde{z}}{a}$ will be the amount of x produced.

Consider now two identical economies, embedded in the world, that face the same P, r, R, etc. with exactly similar distribution of $k(z)$. Suppose y can be produced in fragments i.e. by splitting up the production process, `a la Jones and Kierzkowski (2001), so that if one country produces λ fraction of the value, the other will produce (1- λ) fraction, trade and mix them to produce one unit of y. Let us then rewrite the incentive constraint (1) when the economy, we started with, produces λ fraction of y and imports (1- λ) fraction from the other, from its exactly identical counterpart.

The new constraints are

$$\lambda P - \left[\frac{\lambda}{a} - k(z)\right](1+R) + \frac{1}{a}(1+r) \geq \frac{1}{a}(1+r) + k(z)(1+r) \qquad (5)$$

The country gets λ fraction of the net value employing λ fraction of worker who can work (1- λ) fraction in x and get $\frac{1}{a}(1+r)$ in total and (5) can be rewritten as

$$P + \frac{k(z)}{\lambda}(R-r) \geq \frac{1}{a}\,(1+R) \qquad (6)$$

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Note that the LHS of (6) is greater than the LHS in (2), implying greater incentive to produce y. Figure 1 depicts $\tilde{z}$ in two different cases with $\tilde{z}_2$ (in (6)) being less than $\tilde{z}_1$ (in (2)).

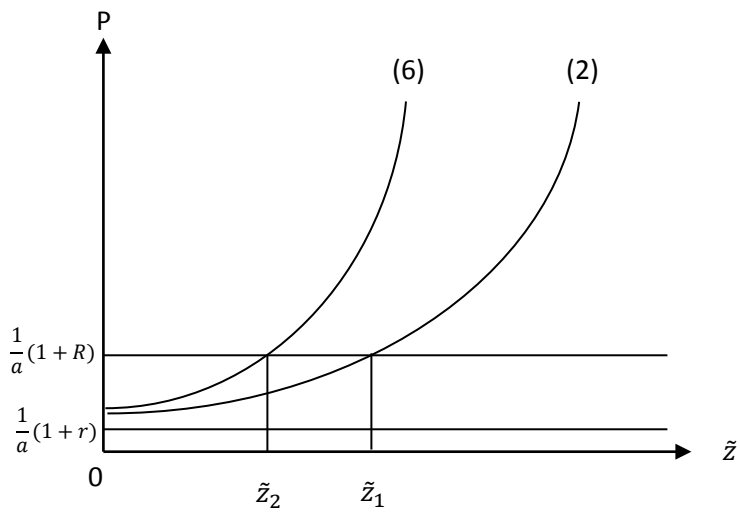

Figure 1

As $\lambda < 1$, LHS in (6) is greater than RHS in (6) for $\tilde{z} = \tilde{z}_1$, implying a drop in $\tilde{z}$ to $\tilde{z}_2$. We have drawn Figure 1 to indicate that $P > \frac{1}{a}(1+r)$ implies that if there is no distortion in the credit market, the economy would have specialized in y. If we allow trade in fragments $\tilde{z}$ drops and we have $\lambda\big((1-\tilde{z}(\lambda)\big)$ working hours being spent on y. Another important point is that the trading nations, both suffering from credit market imperfection, will trade without any reference to comparative advantage. It is possible that the closed economy can engage in internal trade when agents produce fragments of y. But whenever another country appears in the horizon, there will be inter-country trade provided technological considerations do not prohibit further fragmentation. The point we are trying to make is that, since $R > r$, there are latent diseconomies of scale and fragmentation allows higher income. We take up these issues in more detail in the following sections.

III. Trade, Fragmentation and First Best Allocation

How does $\lambda\big((1-\tilde{z}(\lambda)\big)$ behave once we lower λ which leads to a decline in $\tilde{z}$? The following proposition answers this question.

Proposition 1. Trade in Fragments allows a movement towards first best allocation from autarky i.e. $(\lambda = 1)$ iff $\tilde{z} > \frac{1}{1+\frac{1}{\varepsilon}}$ where $\epsilon \equiv \frac{k'(\tilde{z})}{k(\tilde{z})}\tilde{z}$

<u>Proof</u>: If $P > \frac{1}{a}(1+r)$, only y will be produced and we define it as the first best allocation where

$\tilde{z} = 0$, All working hours are devoted to production of y.

Given that $k(\tilde{z}) = \frac{\lambda\left[\frac{1}{a}(1+R)-P\right]}{(R-r)}$ (from (6)),

$$\frac{d\left[\lambda\left(1-\tilde{z}(\lambda)\right)\right]}{d\lambda} = 1 - \tilde{z}\left(1+\frac{1}{\varepsilon}\right) \tag{7}$$

where $\varepsilon = \frac{k'(\tilde{z})}{k(\tilde{z})}\tilde{z} > 0$ denotes the elasticity of distribution of capital or wage fund stock with respect to $\tilde{z}$. Therefore, (7) will imply

$$\frac{d[\lambda(1-\tilde{z}(\lambda)]}{d\lambda} < 0 \text{ iff } \tilde{z} > \frac{1}{1+\frac{1}{\varepsilon}} \tag{8}$$

If, initially (with $\lambda = 1$), $\tilde{z} > \frac{1}{1+\frac{1}{\varepsilon}}$ and we allow trade in fragments $\tilde{z}$ drops and given given (8) L_y i.e. total allocation of labour in y increases and we move towards the first best. (QED)

Proposition 1 clearly states that for $\tilde{z}$ lower than $1+\frac{1}{\varepsilon}$, trade in fragments will induce more people to be entrepreneurs but total allocation of labour for production of y will actually fall. As more people come into sector y, the existing ones allocate less working hours towards production of y. It is a tradeoff between extensive and intensive margin. Countries with smaller amounts of $k(z)\forall\, z$ are likely to generate a higher $\tilde{z}$ at the initial equilibrium $(\lambda = 1)$ implying the

reluctance of the people to move into y as $R > r$. Surely, in such cases trade in fragments will lead to greater production of y and lower production of x. However, if initial $\tilde{z}$ is fairly low, further decline in $\tilde{z}$ may actually reduce production of y. Since $(1-\tilde{z})$ is relatively high, a drop in λ, from $\lambda = 1$, impacts y more heavily which can not be compensated by allowing more workers in production of y.

Let us now consider the possibility that at least some workers are endowed with enough capital so that they do not have to borrow. Since $k^{'}(z) > 0$, let us stipulate some $\bar{z}$ such that $k(\bar{z}) = \frac{1}{a}, \bar{z} < 1$. Therefore, for $z\epsilon[\bar{z}, 1]$, workers-entrepreneurs do not need to borrow.This incentive constraint boils down to

$$\mathrm{P} + \frac{1}{\mathrm{a}}\,(1+\mathrm{r}) + \left(\mathrm{k(z)} - \frac{1}{\mathrm{a}}\right)(1+\mathrm{r}) \geq \frac{1}{\mathrm{a}}(1+\mathrm{r}) + \mathrm{k(z)}(1+\mathrm{r})$$

$$or, P \geq \frac{1}{a}(1+r)$$

In Figure 1, P exceeded $\frac{1}{a}(1+r)$ and, therefore, all such worker-entrepreneurs beyond $\bar{z}$ would definitely produce y. Figure 2 summarizes the new equilibrium.

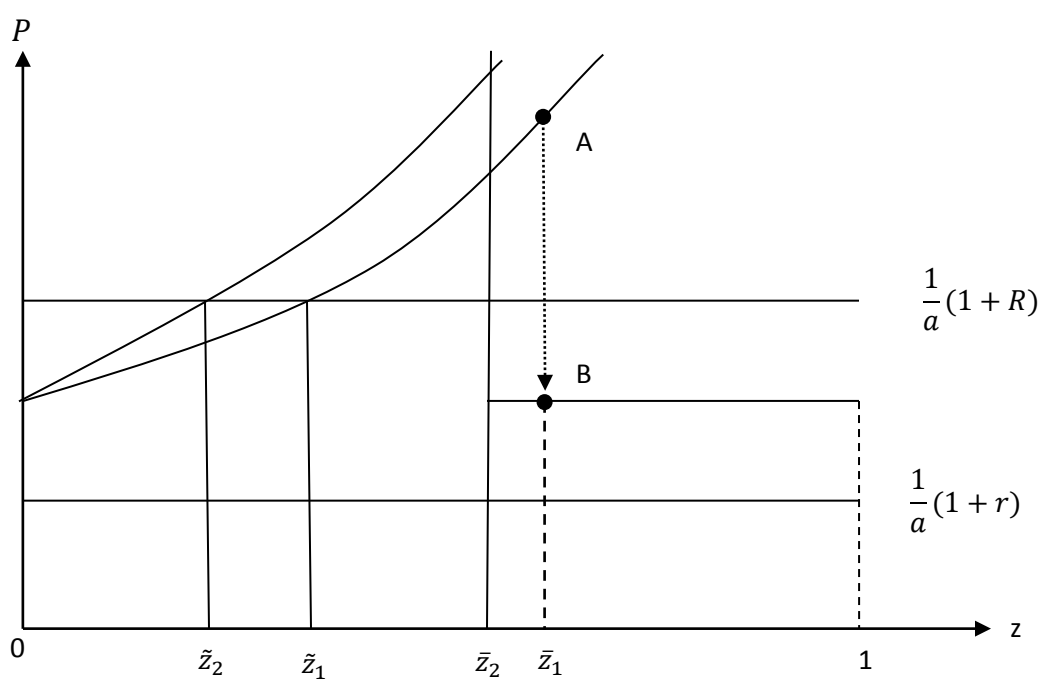

Figure 2

Initially workers from 0 to $\tilde{z}_1$ were engaged in x and $\tilde{z}_1$ to 1 were engaged in y. But for $z \in (\bar{z}_1, 1)$, the incentive constraint jumps down from A to B and coincides with the price line. With fragmentation $\tilde{z}_1$ declines up to $\tilde{z}_2$ and $z_1$declines up to $\bar{z}_2$.

In fact, relatively rich workers i.e. indexed $\bar{z}$ and beyond do not need to engage in trade in fragments since λ does not affect their incentive constraint. However, now more people do not face the constraint any more and $\bar{z}_2 < \tilde{z}_1$. Thus the prediction of the model is that with credit constraints binding, relatively poor will engage in trade. Net benefit from trade, net of the opportunity cost is greater for those who face the credit constraint because for them having greater amount of internal finance provides an added benefit. For those who does not face the borrowing constraint own finances do not matter. While relatively rich will be reluctant to trade in fragments, relatively poor will be eager to engage in such a trade. Of course those who continue to produce x will not derive the benefit from trade in fragments. Thus the workers are divided into three categories. First, the poorest who wish to move into y sector but will find credit too costly to borrow and will produce x. Second, those who will borrow and trade in fragments to produce y. Third, the richest ones who will produce y but will be indifferent towards trade in fragments.

IV. General Equilibrium with endogenous P

In the preceding section we pinned down $\tilde{z}$ by

$$P + k(\tilde{z})(R - r) = \frac{1}{a}(1 + R) \tag{8}$$

Given k'>0, (8) implies that with an increase in P, $k(\tilde{z})$ will fall and $\tilde{z}$ will fall. Total production of Y denoted by (1 - $\tilde{z}$) will rise and as $\tilde{z}$ goes down, production of X will decline. Hence,

$$Y/_X = f(P) \tag{9}$$

where $f' > 0$.

We close the model by a homothetic demand function

$$\frac{D_Y}{D_X} = \Phi(P) \tag{10}$$

where $\Phi' < 0$.

From (2) and (3) we determine the autarkic equilibrium relative price P_0 and the rest of the variables given R and r. Note that we shall get identical relative price in both countries as they are exact replica of each other.

As we open up trade in fragments we get different $\tilde{z}$ in both countries. Without loss of generality let us assume that the production process of Y can be divided into only two fragments of equal size. Such a division is technologically given which splits up the process in half.

The way we have defined $\tilde{z}$ now is

$$P + \frac{k(\tilde{z})}{1/2}(R - r) = 1/a\,(1 + R) \tag{11}$$

We have shown earlier that at any given P, $\tilde{z}$ will fall. Given P, we have same $\tilde{z}$ in both countries and they are lower. Total production of Y at a given P in each country is determined by

$$Y = \frac{1}{2}(1 - \tilde{z}_T) \tag{12}$$

Total production in each country will increase iff the following holds

$$\tilde{z}_T < 2\tilde{z} - 1 \tag{13}$$

The condition above states that if the new $\tilde{z}$ is lower than the old one by a certain margin, total production in each country will increase and P will fall if we allow fragmentation. The significance of this result can be seen as follows. In the standard literature in industrial organization and financial economics (Nocke and Thanassoulis (2009)) credit constraints, that force the firms to choose smaller size or fragment, tend to increase price for the consumers due

to the problem of double marginalization. In this set up we can have a rising P as a consequence of fragmentation if we ignore the effect of fresh entry into the business by workers-entrepreneurs. But if we allow the entry effect to be significant fragmentation must reduce price as well and improve efficiency. Of course market structure does matter and efficiency of vertical relation will depend on price setting capacity of producers in an imperfectly competitive set up. But in that set up also one cannot ignore the consequent entry of firms who now find it profitable to engage in the production of Y.

V. Concluding Remarks

The main message of our paper is that economies constrained by the availability of credit, as reflected in a difference between the borrowing and the lending rate, will have a natural tendency to trade with each other by fragmenting the production process. Thus, identical countries will gain from trade by sharing production of the credit intensive good releasing the pressure on their internal finance. In the typical Dixit-Stiglitz-Krugman class of models, identical economies gain from trade by consuming greater number of varieties. In comparison, ours is Ricardian model of inter industry trade where comparative advantage does not play any role as countries trade even if they have exactly identical technologies. Smaller sized activities to some extent alleviate the problem of credit market imperfection. International trade in fragments allows firms to choose smaller size of output which suffers the credit constraint. Our results suggest that those with more than adequate amount of capital are not likely to engage in such trade simply because internal financing does not assume a special role in this context. Relatively poor workers – entrepreneurs will engage in trade in intermediate or fragments. Fragmentation contributes in terms of encouraging entrepreneurs among the relatively poor echoing the observations in Jones and Marjit (2001).

In this paper, we have abstracted from any differences between the countries that can lead to comparative advantage. It does not matter *how* production activity is divided between countries: as long as such activities *can* be divided, there will be natural gains from trade. In an otherwise scale neutral world, availability of credit assumes a critical role. "Small" as a size seems to be more convenient than the "Large". We have not explicitly worked out the case when a country with unconstrained entrepreneurs faces another where credit is available only at a premium. It is clear that as long as $P > 1/a(1+r)$, the richer nation will specialize in Y, but, in that case, the constrained economy will do better in supplying fragments to the richer nation. Some interesting extensions of our model may involve allowing product differentiation[i], including public firms[ii], introducing costly technology transfers[iii], incorporating costs of complexity[iv], or injecting urban unemployment[v].

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[i] See Batra (1972, 1974), Khan (1975), and Khan and Vohra (1985).
[ii] See Chao and Yu (2006).
[iii] See Jones and Ruffin (2008).
[iv] See Gale and Sabourian (2005).
[v] See Chau and Khan (2001) and Oladi and Gilbert (2011).